

The empirical recurrence for OEIS A298967: recovering a conjecture that is not written down

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Abstract

OEIS A298967 records “Empirical recurrence of order 75”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 1024$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 75, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 75 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A298967 is “Number of $n \times 5$ 0..1 arrays with every element equal to 0, 1, 2, 3, 4 or 5 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

16, 512, 11283, 280468, 6892031, 170137416, 4200575252, 103715870545, ...

The entry states:

Empirical recurrence of order 75 (see link above).

As of the “Last modified” line on the live entry (Jan 01 2024 11:51:48, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1024$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 1024$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 2048$ exact terms of the model returns order 75 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{75} c_i a(n-i),$$

c_1	23	c_{20}	577607041890	c_{39}	249009028996704	c_{58}	13101426422089
c_2	73	c_{21}	1076019010690	c_{40}	679911373312013	c_{59}	10371501996931
c_3	−537	c_{22}	−1431802655239	c_{41}	50760637363472	c_{60}	−1478818680803
c_4	−5845	c_{23}	−5910740798273	c_{42}	−747699088519801	c_{61}	−3569930810127
c_5	−4522	c_{24}	−1023587659184	c_{43}	−451039453487158	c_{62}	−411714480701
c_6	138614	c_{25}	18730534312245	c_{44}	483200522047931	c_{63}	943936749641
c_7	355996	c_{26}	23253112865849	c_{45}	629301372722330	c_{64}	300479032894
c_8	−1461206	c_{27}	−26068761719363	c_{46}	−102819953151283	c_{65}	−165747216434
c_9	−8093388	c_{28}	−83635615856664	c_{47}	−507170153802124	c_{66}	−118113537196
c_{10}	5967959	c_{29}	−27143898751207	c_{48}	−117293305916518	c_{67}	−12122231208
c_{11}	104620727	c_{30}	137536624241402	c_{49}	289428431103856	c_{68}	13049902064
c_{12}	87523367	c_{31}	169561096119549	c_{50}	154491360484736	c_{69}	6203489392
c_{13}	−868518296	c_{32}	−78327790285156	c_{51}	−136559639554651	c_{70}	1498066816
c_{14}	−1981829306	c_{33}	−297201657238687	c_{52}	−129255964418247	c_{71}	−362216320
c_{15}	4480108770	c_{34}	−83285791673219	c_{53}	45227129545929	c_{72}	−158977536
c_{16}	19863954397	c_{35}	322586867378088	c_{54}	88296130020901	c_{73}	128108544
c_{17}	−7076711973	c_{36}	235777614405662	c_{55}	3647682790823	c_{74}	25214976
c_{18}	−129757664328	c_{37}	−320243229641131	c_{56}	−43426946831743	c_{75}	−5455872
c_{19}	−99347597033	c_{38}	−427111361948495	c_{57}	−16150027907687		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 75, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $75 + 4$ terms removed returns order 75 as well, so $\deg q_{\min} = 75 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $75 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 75 that this sequence satisfies — and that family turns out to have exactly one member.

References

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